



Centre for Development of Advanced Computing (C-DAC)

Written Examination- Technical Assistant

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POST CODE: - CDACS/TA/NetOps/01

Domain- Computer Network & System Administration & Network Security

Name of the Candidate:

Position Applied For: Technical Assistant.

Domain:

Candidate ID:

Date of Examination:

.....
(Signature of Invigilator)

.....
(Signature of the Candidate)

General Instruction

1. No Electronic Gadgets including Mobile Phone (even in switched off condition) are allowed inside the Examination Hall.
2. Ball Pen (Blue/Black), Admit Card and Photo ID is allowed inside the examination hall.
3. For each correct answer 2 marks will be awarded and for each wrong answer 1 mark shall be deducted.
4. Rough Work can be done on rough sheets provided and no damage/ tearing or folding or doing rough work in OMR Sheet shall be acceptable. If found so, candidature is liable to be cancelled.
5. It is to be ensured that the OMR sheet is signed by candidate and countersigned by the invigilator before handing over the OMR sheet.
6. The Question Paper has two parts, Part A (English & Aptitude) of 20 Questions and Part B (Domain Knowledge) of 80 Questions.

PART- A

Directions (1-2): There is a sentence given below which is divided into four or five parts. There is an error in any part(s) of the sentence. From the given options, find out the part which contain(s) error.

1) When opposing political parties both think they are winning (A)/, one of them are usually wrong and this is the quandary (B)/America finds itself in (C), after the Democrats won the House of Representatives on the suburbs (D) , while the Republicans tightened their grip on the Senate in the sticks

- A) Only A and B B) Only A and C
C) Only B and D D) Only A and D

2) Toxic federal politics is America great weakness (A) / and it prevents action on pressing real issues (B)/, from immigration to welfare; it erodes Americans' faith (C)/of their government and its institutions; and (D) /it dims the beacon of American democracy abroad.

- A) Only A and B B) Only A and C
C) Only B and D D) Only A and D

Direction (3): Four words in each sentence have been highlighted. They may or may not be in their correct positions. One of these four words may be incorrect in the context. Choose the option which provides the pair of words to be interchanged to make the sentence meaningful along with the appropriate word that replaces the wrong one.

3) It is worth remembering that the recently and soon-to-be retired have already seen large cuts to their **(A) expectation** state pension income due to **(B) increases** in state pension age – which impact most **(C) unjustly** upon poorer groups with lower life **(D) lifetime**

- A) B-C; increase B) B-D; no change
C) A-B; increase D) A-D; expectancy

Directions (4-5): In this question, a part of the sentence is made bold. Below are given alternatives to the bold part at (1), (2), (3) and (4) which may improve the sentence. Choose the correct alternative.

4) The video has already gathered nearly 40,000 views – **and** the numbers are only increasing.

- A) because B) but C) yet D) so

5) Do you follow the different social media profiles managed by Nasa? Then you may have seen the sonification clips **that translate data's** from outer space into sound.

- A) those translate datas
B) that translate data
C) this translate datas
D) these translate data's

6) Choose the synonym of **Driver**.

- A) Straphanger B) Catalyst
C) Prototype D) Chauffer

7) Choose the antonym of **Hinder**

- A) Help B) Host
C) Unhealthy D) Unfamiliar

8) The missing term in the sequence PQR, P²QR, P²Q²R.....P³Q²R² is

- A) P³Q²R B) P²Q²R²
C) P³Q³R² D) P³QR

9) If CANE is coded as 1345 and MEAN as 8453, then TOKENS can be coded as

- A) 765239 B) 142530
C) 764539 D) 762039

10) A train of length 250 meter long is running at the speed of 10 m/s. How much time it will take to cross a platform of 650 meter length,

- A) 60 seconds B) 72 seconds
C) 85 seconds D) 90 seconds

11) The monthly income of A and B are in the ratio 2:3 and their monthly expenses are in the ratio 5:9. If each of them saves Rs 600 per month, then their monthly incomes are

- A) Rs 1600, Rs 2400
B) Rs 1650, Rs 2200
C) Rs 1800, Rs 2600
D) Rs 1950, Rs 2750

12) A wheel makes 1000 rotation in covering a distance of 44 km. Find the radius of the wheel

- A) 5.5 m B) 6 m C) 7 m D) 7.5 m

13) If $5^a = 3125$, then the value of $5^{(a-3)}$ is

- A) 25 B) 125 C) 625 D) 1625

14) If the mathematical operator $\div$ means $\times$, $+$ means $-$, $\times$ means $+$ and $-$ means $\div$, then $25 + 18 - 3 \times 7 \div 3 = ?$

- A) 25 B) 21 C) 19 D) 40

15) Which two signs (mathematical operators) should be interchanged to make the given equation correct?

$$56 - 8 + 42 \div 6 \times 5 = 19$$

- A) $-$ and $\times$ B) $+$ and $-$
C) $\div$ and $-$ D) $+$ and $\div$

16) If 'P3Q' means P is the daughter of Q, 'P5Q' means P is the father of Q, 'P7Q' means P is the mother of Q, and P9Q means P is the sister of Q, then how is J related to K in 'J3L9N3M5K'?

- A) Mother
- B) Wife
- C) Niece
- D) Daughter

17) Odometer is to mileage as compass is to

- A) Speed
- B) Hiking
- C) Needle
- D) Direction

18) Excluding stoppage, the speed of a bus is 54 km/hr and including stoppage, it is 45 km/hr. For how many minutes the bus stops per hour?

- A) 9
- B) 10
- C) 12
- D) 20

19) If $\log x + \log y = \log (x + y)$, then

- A) $x = y$
- B) $xy = 1$
- C) $y = (x-1) / x$
- D) $y = x / (x-1)$

20) John sees a train passing over 1 km long bridge. The length of the train is half that of the bridge. If the train clears the bridge in 2 min, find the speed of the train

- A) 45kph
- B) 43kph
- C) 50kph
- D) None

PART B

21) How long is an IPv6 address?

- a) 32 bits b) 128 bytes
- c) 64 bits d) 128 bits

22) Which of the following is the valid host range for the subnet on which the IP address 192.168.168.188 255.255.255.192 resides?

- a) 192.168.168.129-190
- b) 192.168.168.129-191
- c) 192.168.168.128-190
- d). 192.168.168.128-192

23) Which protocol does DHCP use at the Transport layer?

- a) IP b) TCP c) UDP d) ARP

24) Where is a hub specified in the OSI model?

- a) Session layer b) Physical layer
- c) Data Link layer d) Application layer

25) Which class of IP address has the most host addresses available by default?

- a) A b) B c) C d) A and B

26) What is the main reason the OSI model was created?

- a) To create a layered model larger than the DoD model.
- b) So application developers can change only one layer's protocols at a time.
- c) So different networks could communicate.
- d) So Cisco could use the model.

27. What layer in the TCP/IP stack is equivalent to the Transport layer of the OSI model?

- a) Application b) Host-to-Host
- c) Internet d) Network Access

28. Which of the following is private IP address?

- a) 12.0.0.1 b) 168.172.19.39
- c) 172.15.14.36 d) 192.168.24.43

29. If you use either Telnet or FTP, which is the highest layer you are using to transmit data?

- a). Application b). Presentation
- c) Session d). Transport

30 Which of the following describe router functions?

- a). Packet switching
- b). Packet filtering
- c). Internetwork communication
- d). All of the above

31. What are the decimal and hexadecimal equivalents for the binary number

10110111?

- a) 69/0x2102 b) 183/B7
- c) 173/A6 d) 83/0xC5

32. What is the purpose of flow control?

- a) To ensure that data is retransmitted if an acknowledgment is not received.
- b) To reassemble segments in the correct order at the destination device.
- c) To provide a means for the receiver to govern the amount of data sent by the sender.
- d) To regulate the size of each segment.

33. Segmentation of a data stream happens at which layer of the OSI model?

- a) Physical b) Data Link
- c) Network d) Transport

34. What type of RJ45 UTP cable is used between switches?

- a) Straight-through
- b) Crossover cable
- c) Crossover with a CSU/DSU
- d) Crossover with a router in between the two switches

35. What is the maximum number of IP addresses that can be assigned to hosts on a local subnet that uses the 255.255.255.224 subnet mask?

- a. 14 b. 15
- c. 16 d. 30

36. Default port number used by SSH and telnet are respectively

- a) 22, 23
- b) 23, 20
- c) 20, 21
- d) 21,20

37. A DNS client is called

- a) DNS updater
- b) DNS resolver
- c) DNS handler
- d) none of the mentioned

38. Servers handle requests for other domains _

- a) directly
- b) by contacting remote DNS server
- c) it is not possible
- d) none of the mentioned

39. If a server has no clue about where to find the address for a hostname then

- a) server asks to the root server
- b) server asks to its adjacent server
- c) request is not processed
- d) none of the mentioned

40. Which one of the following allows client to update their DNS entry as their IP address change?

- a) dynamic DNS
- b) mail transfer agent
- c) authoritative name server
- d) none of the mentioned

41. The domain name system is maintained by _

- a) distributed database system
- b) a single server
- c) a single computer
- d) none of the mentioned

42. NTP time servers work within the TCP/IP suite and rely on -

- a) TCP port 121 b) UDP port 121
- c) TCP port 123 d) UDP port 123

43. A sample firewall configuration of a Linux webserver is shown below:

```
<forward-port to-port="80" protocol="tcp" port="8080"/>
```

What does the above firewall rule signify?

- a) Webserver is listening on port 80.
- b) Web application is listening on port 8080.
- c) Web application is listening on port 80
- d) Both A & B

44. The physical layer translates logical communication requests from the_____

_____ into hardware specific operations.

- a) Data link layer b) Network layer
- c) Transport layer d) Application layer

45. Primary reasons to deploy RAID by System Administrators include:

- a). Enhances speed
- b) Increases storage capacity using a single virtual disk
- c) Minimizes disk failure d) All of these

46. Which sublayer of the data link layer performs data link functions that depend upon the type of medium?

- a) logical link control sublayer
- b) media access control sublayer
- c) network interface control sublayer
- d) error control sublayer

47. Header of a frame generally contains

- a) synchronization bytes
- b) addresses
- c) frame identifier
- d) all of the mentioned

48. Automatic repeat request error management mechanism is provided by

- a) logical link control sublayer
- b) media access control sublayer
- c) network interface control sublayer
- d) application access control sublayer

49. When 2 or more bits in a data unit has been changed during the transmission, the error is called _

- a) Random error b) Burst error
- c) Inverted error d) Double error

50. CRC stands for

- a) cyclic redundancy check
- b) code repeat check
- c) code redundancy check
- d) cyclic repeat check

51. The technique of temporarily delaying outgoing acknowledgements so that they can be hooked onto the next outgoing data frame is called _

- a) piggybacking
- b) cyclic redundancy check
- c) fletcher's checksum
- d) parity check

52. Web server refers to -

- a) Hardware b) Software
- c) Hardware or Software, or combination of both
- d) None of these.

53. FTP is built on _____ architecture.

- a) Client-server
- b) P2P
- c) Data centric
- d) Service oriented

54. Two devices are in network if

- a) a process in one device is able to exchange information with a process in another device
- b) a process is running on both devices
- c) PIDs of the processes running of different devices are same
- d) a process is active and another is inactive

55. What are the decimal and hexadecimal equivalents of the following binary number 110110111?

- a) 439 and 1B7
- b) 539 and 2B7
- c) 339 and 1A7
- d) 934 and 1B7

56. In wireless ad-hoc network

- a) access point is not required
- b) access point is must
- c) nodes are not required
- d) all nodes are access points

57. A is a device that forwards packets between networks by processing the routing information included in the packet.

- a) Bridge b) Firewall
- c) Router d) Hub

58. A list of protocols used by a system, one protocol per layer, is called

- a) protocol architecture
- b) protocol stack
- c) protocol suite
- d) protocol system

59. Network congestion occurs

- a) in case of traffic overloading
- b) when a system terminates
- c) when connection between two nodes terminates
- d) in case of transfer failure

60. The HTTP request message is sent in part of three-way handshake.

- a) First b) Second
- c) Third d) Fourth

61. In the process of fetching a web page from a server the HTTP request/response takes RTTs.

- a) 2 b) 1
- c) 4 d) 3

62. The first line of HTTP request message is called

- a) Request line
- b) Header line
- c) Status line
- d) Entity line

63. The method when used in the method field, leaves entity body empty.

- a) POST b) SEND
- c) GET d) PUT

64. Following command prints:

`tcpdump 'icmp[icmptype] != icmp-echo and icmp[icmptype] != icmp-echoreply'`

- a) All ICMP packets
- b) Only ping packets
- c) All ICMP packets that are echo requests/replies
- d) All ICMP packets other than ping packets

65. Which of the following is present in both an HTTP request line and a status line?

- a) HTTP version number
- b) URL
- c) Method
- d) None of the mentioned

66. MPLS (Multiprotocol Label Switching) works in

- a) Layer
- b) Layer 4
- c) Between Layer 3 and Layer 4
- d) None of these

67. Expansion of SMTP is

- a) Simple Mail Transfer Protocol
- b) Simple Message Transfer Protocol
- c) Simple Mail Transmission Protocol
- d) Simple Message Transmission Protocol

68. By default, how many days does vSphere keep information about tasks and events?

- a) 60 b) 14
- c) 30 d) 7

69. A 4 byte IP address consists of

- a) only network address
- b) only host address
- c) network address & host address
- d) network address & MAC address

70. Which of the following routing algorithms can be used for network layer design?

- a) shortest path algorithm
- b) distance vector routing
- c) link state routing
- d) all of the mentioned

71. Which of the following is not correct in relation to multi-destination routing?

- a) is same as broadcast routing
- b) contains the list of all destinations
- c) data is not sent by packets
- d) there are multiple receivers

72. The network layer protocol for internet is

- a) Ethernet b) Internet protocol
- c) hypertext transfer protocol
- d) file transfer protocol

73. ICMP is primarily used for _

- a) error and diagnostic functions
- b) addressing
- c) forwarding
- d) routing

74. Physical or logical arrangement of network is

- a) Topology
- b) Routing
- c) Networking
- d) Control

75. Which network topology requires a central controller or hub?

- a) Star b) Mesh
- c) Ring d) Bus

76. Data communication system within a building or campus is

- a) LAN b) WAN
- c) MAN d) PAN

77. In TDM, slots are further divided into _

- a) Seconds b) Frames
- c) Packets d) Bits

78. The PPP protocol

- a) Is designed for simple links which transport packets between two peers
- b) Is one of the protocols for making an Internet connection over a phone line
- c) Is designed for simple links which transport packets between two peers and making an Internet connection over a phone line
- d) Is used for sharing bandwidth

79. The PPP encapsulation

- a) Provides for multiplexing of different network-layer protocols
- b) Requires framing to indicate the beginning and end of the encapsulation
- c) Establishing, configuring and testing the data-link connection
- d) Provides interface for handling the capabilities of the connection/link on the network

80. A network manager wishes to assign static IP addresses to the servers in a network. For network 10.10.10.24/29, the router is assigned the first usable host address while the Web server is assigned the last usable host address. Which IP configuration values should be assigned to the Web server?

a) IP address: 10.10.10.14 Mask: 255.255.255.248 Gateway: 10.10.10.9

b) IP address: 10.10.10.254 Mask: 255.255.255.0 Gateway: 10.10.10.1

c) IP address: 10.10.10.30 Subnet Mask: 255.255.255.240 Gateway: 10.10.10.17

d) IP address: 10.10.10.30 Mask: 255.255.255.248 Gateway: 10.10.10.25

81. The attacker using a network of compromised devices is known as

- a) Internet
- b) Botnet
- c) Telnet
- d) D-net

82. Which of the following is a form of DoS attack?

- a) Vulnerability attack
- b) Bandwidth flooding
- c) Connection flooding
- d) All of the mentioned

83. Firewalls are often configured to block

- a) UDP traffic
- b) TCP traffic
- c) Sensitive traffic
- d) Best-effort traffic

84. In a network, If P is the only packet being transmitted and there was no earlier transmission, which of the following delays could be zero?

- a) Propagation delay
- b) Queuing delay
- c) Transmission delay
- d) Processing delay

85. Coaxial cable consists of concentric copper conductors.

- | | |
|------|------|
| a) 1 | b) 2 |
| c) 3 | d) 4 |

86. When a ping to the local host IP address succeeds but a ping to the default gateway IP address fails, what can you rule out?

- a) The IP address of the local host is incorrect.
- b) The IP address of the gateway is incorrect.
- c) The IP stack has failed to initialize on the local system.
- d) The NIC is not functional on the gateway.

87. You have a device with a MAC address of c7-c6-76-4d-44-60. What will be the automatically generated link-local address?

- a) fe80::c5c6:76ff:fe4d:4460
- b) fe80::c5c6:764d:4460
- c) 2001::c5c6:76ff:fe4d:4460
- d) fe80::c7c6:76ff:fe4d:4460

88. In cryptography, what is cipher?

- a) algorithm for performing encryption and decryption
- b) encrypted message
- c) both algorithm for performing encryption and decryption and encrypted message
- d) decrypted message

89. Cryptanalysis is used

- a) to find some insecurity in a cryptographic scheme
- b) to increase the speed
- c) to encrypt the data
- d) to make new ciphers

90. An organization with 40 clients having an ILL connection of 20 Mbps with sharing ratio 1:4. The organization wants to host its own website (public) encrypted using Transport Layer Security (TLS) in the De-Militarized Zone. Considering the default rule is reject, the appropriate rule that needs to be applied in inbound and outbound connection is

- a) Inbound: allow 80, 443 & Outbound: allow 80, 443
- b) Inbound: allow 443
- c) Inbound: allow 53, 443,80 & Outbound: allow 443
- d) Outbound: allow 443

91. DHCP (dynamic host configuration protocol) provides to the client.

- a) IP address b) MAC address
- c) Url d) None

92. Telnet protocol is used to establish a connection to

- a) TCP port number 21
- b) TCP port number 22
- c) TCP port number 23
- d) TCP port number 25

93. Bluetooth is the wireless technology for

- a) local area network
- b) personal area network
- c) metropolitan area network
- d) wide area network

94. Unauthorised access of information from a wireless device through a bluetooth connection is called

- a) bluemarking b) bluesnarfing
- c) bluestring d) bluescoping

95. Which of the following computer networks is built on the top of another network?

- a) prior network b) chief network
- c) prime network d) overlay network

96. Communication channel is shared by all the machines on the network in

- a) broadcast network
- b) unicast network
- c) multicast network
- d) anycast network

97. What ports should be blocked on the firewall to prevent NetBIOS traffic from not coming through the firewall if your network is comprised of Windows NT, 2000, and XP? (Choose two)

- a) 139 b) 110
- c) 445 d) 80

98. TCP/IP model was developed the OSI model.

- a) Prior to b) After
- c) Simultaneous to d) With no link to

99. What is the runlevel to halt (shut down) the UNIX system.

- a) init 0 b) init 1
- c) init 2 d) init 3

100. A high-performance journaling file system that supports maximum file size of 8 EiB is

- a) JFS b) NTFS
- c) ext4 d) XFS